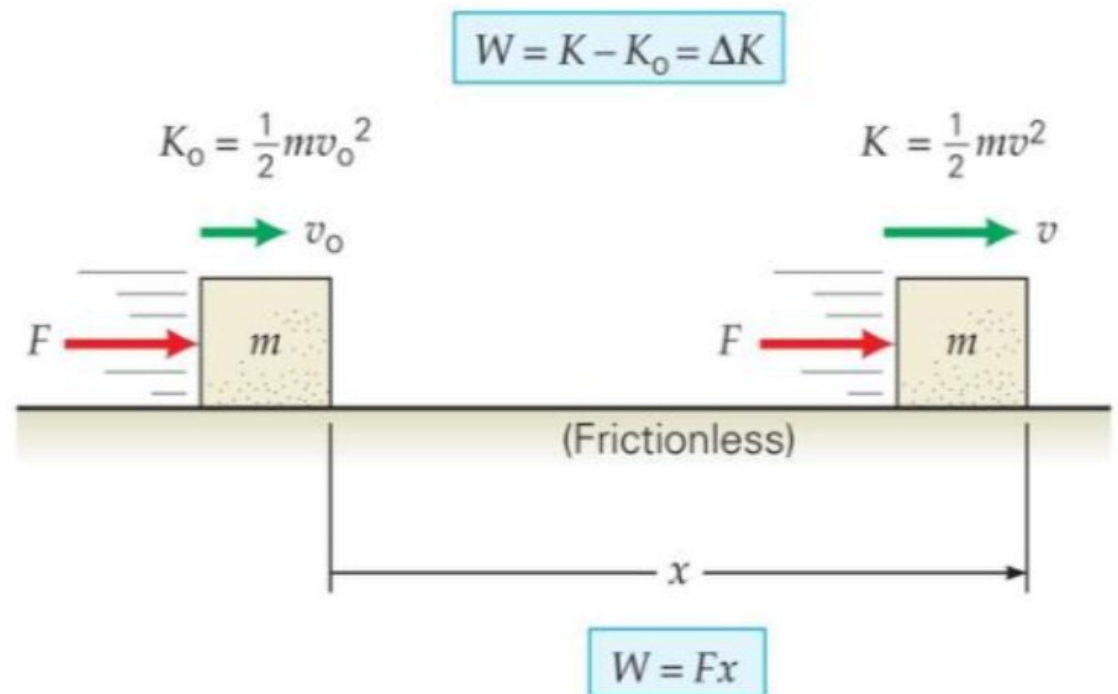


The Work-Energy Theorem: Kinetic Energy

We'll start with the form of energy that is most closely related to work – **kinetic energy (KE)**.

Consider the figure below, in which an object of mass m has an initial speed v_0 . A constant force F acts on the object (in the same direction as its velocity), increasing the speed to v . The displacement of the object is x .

From the last lecture, we know that the work done on the object is $W = Fx$. Then, from chapter 2, we have $v^2 = v_0^2 + 2ax$, which implies that the object has acceleration $a = \frac{v^2 - v_0^2}{2x}$.



Work-Kinetic Energy Theorem: Kinetic Energy

Now, we apply Newton's second law:

$$F = ma = m \left(\frac{v^2 - v_0^2}{2x} \right)$$

Finally, the work done is

$$W = Fx = m \left(\frac{v^2 - v_0^2}{2x} \right) x = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2.$$

We now define the **kinetic energy (K)** of the object:

$$K = \frac{1}{2}mv^2$$

By this definition, the work done on the object is simply equal to the change in the object's kinetic energy:

$$W = K - K_0 = \Delta K$$

Work-Kinetic Energy Theorem: Kinetic Energy

Kinetic energy can be thought of as **energy possessed by mass that has motion**. It has SI units of Joules, and can not have a negative value (since both m and v^2 are either positive or zero).

The last equation on the previous slide represents the **work-kinetic energy theorem**:

the net work done on a body by all the forces acting on it is equal to the change in kinetic energy of the body

When positive net work is done on an object, its kinetic energy increases (which means its speed increases, assuming that mass is constant). When negative net work is done, the object's kinetic energy and speed must decrease. If net work is zero, the object's KE and speed do not change.

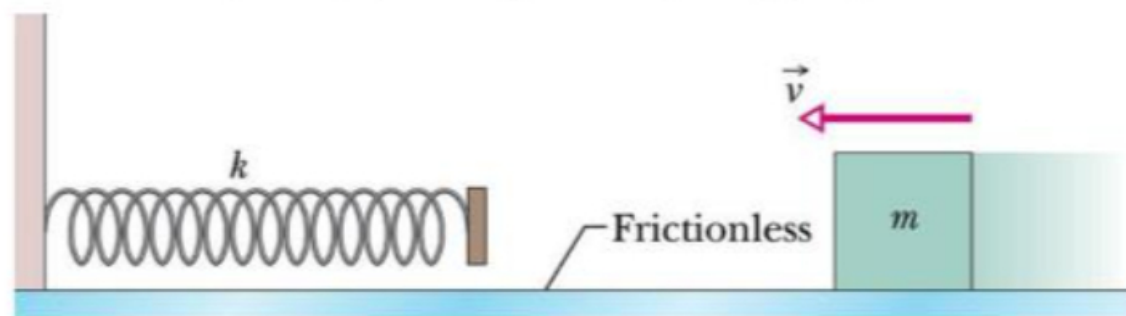
A father racing his son has half the kinetic energy of the son, who has half the mass of the father. The father speeds up by 1.0 m/s and then has the same kinetic energy as the son. What are the initial speeds of both the father and the son?

A constant net force of 75 N acts on an object initially at rest as it moves through a parallel distance of 0.60 m.

- a) What is the final kinetic energy of the object?**
- b) If the object has a mass of 0.20 kg, what is its final speed?**

An out-of-control truck with a mass of 5000 kg is traveling at 35.0 m/s when it starts descending a steep 15° incline with locked brakes. The incline is icy, so the coefficient of friction is only 0.30. Use the work-KE theorem to determine how far the truck will skid before it comes to rest.

A container of mass $m = 0.40$ kg slides across a horizontal, frictionless counter with speed $v = 0.50$ m/s. It runs into and compresses a spring of spring constant $k = 750$ N/m. When the container is momentarily stopped by the spring, by what distance d is the spring compressed?



Chapter 8

Potential Energy and Conservation of Energy

In this chapter we will introduce the following concepts:

Potential energy

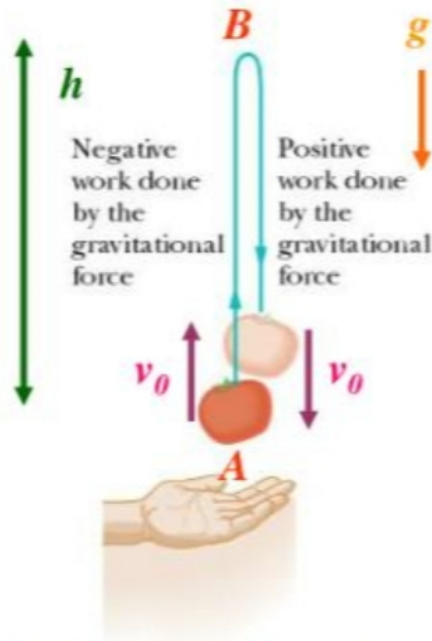
Conservative and nonconservative forces

Mechanical energy

Conservation of mechanical energy

The conservation of energy theorem will be used to solve a variety of problems.

As was done in Chapter 7, we use scalars such as work, kinetic energy, and mechanical energy rather than vectors. Therefore the approach is mathematically simpler.



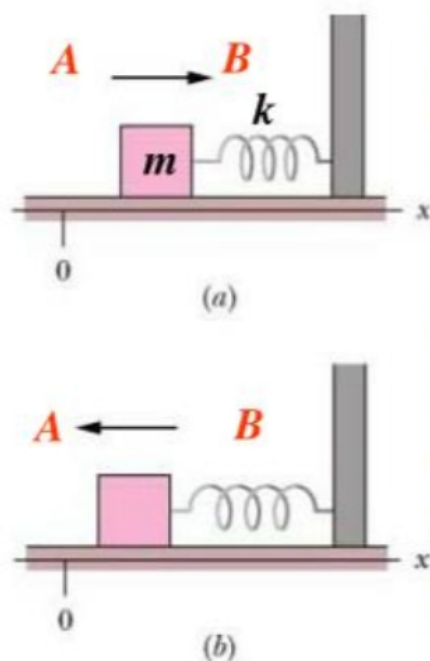
Work and Potential Energy:

Consider the tomato of mass m shown in the figure. The tomato is taken together with the Earth as the system we wish to study. The tomato is thrown upward with initial speed v_0 at point A. Under the action of the gravitational force it slows down and stops completely at point B. Then the tomato falls back and by the time it reaches point A its speed has reached the original value v_0 .

Below we analyze in detail what happens to the tomato-Earth system. During the trip from A to B the gravitational force F_g does negative work $W_1 = -mgh$. Energy is transferred by F_g **from** the kinetic energy of the tomato **to** the gravitational potential energy U of the tomato-Earth system. During the trip from B to A the transfer is reversed. The work W_2 done by F_g is positive ($W_2 = mgh$). The gravitational force transfers energy **from** the gravitational potential energy U of the tomato-Earth system **to** the kinetic energy of the tomato. The change in the potential energy U is defined as

$$\Delta U = -W$$

(8-2)



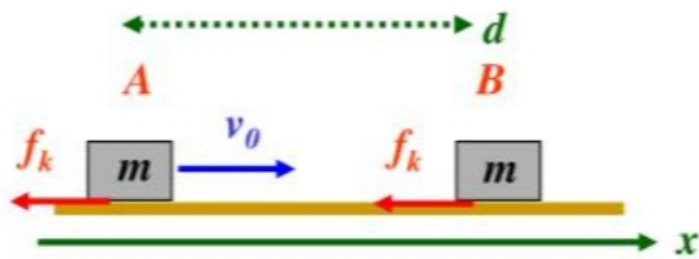
Consider the mass m attached to a spring of spring constant k as shown in the figure. The mass is taken together with the spring as the system we wish to study. The mass is given an initial speed v_0 at point A. Under the action of the spring force it slows down and stops completely at point B, which corresponds to a spring compression x . Then the mass reverses the direction of its motion and by the time it reaches point A its speed has reached the original value v_0 .

As in the previous example, we analyze in detail what happens to the mass-spring system. During the trip from A to B the spring force F_s does negative work $W_1 = -kx^2/2$. Energy is transferred by F_s **from** the kinetic energy of the mass **to** the potential energy U of the mass-spring system.

During the trip from B to A the transfer is reversed. The work W_2 done by F_s is positive ($W_2 = kx^2/2$). The spring force transfers energy **from** the potential energy U of the mass-spring system **to** the kinetic energy of the mass. The change in the potential energy U is defined as

$$\Delta U = -W$$

(8-3)



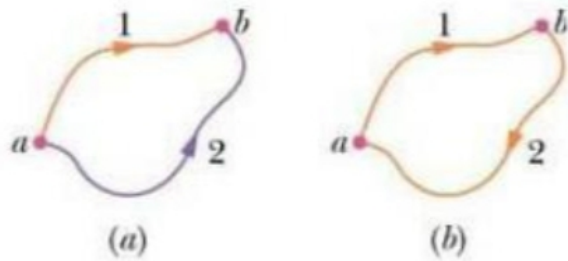
Conservative and Nonconservative Forces.

The gravitational force and the spring force are called “**conservative**” because they can transfer energy from the kinetic energy of part of the system to potential energy and vice versa.

Frictional and drag forces on the other hand are called “**nonconservative**” for reasons that are explained below.

Consider a system that consists of a block of mass m and the floor on which it rests. The block starts to move on a horizontal floor with initial speed v_0 at point A. The coefficient of kinetic friction between the floor and the block is μ_k . The block will slow down by the kinetic friction f_k and will stop at point B after it has traveled a distance d . During the trip from point A to point B the frictional force has done work $W_f = -\mu_k mgd$. The frictional force transfers energy from the kinetic energy of the block to a type of energy called **thermal** energy. This energy transfer **cannot be reversed**. Thermal energy cannot be transferred back to kinetic energy of the block by the kinetic friction. This is the hallmark of nonconservative forces.

(8-4)



Path Independence of Conservative Forces

In this section we will give a test that will help us decide whether a force is conservative or nonconservative.

A force is conservative if the net work done on a particle during a round trip is always equal to zero (see fig. b).

$$W_{\text{net}} = 0$$

Such a round trip along a closed path is shown in fig.b. In the examples of the tomato-Earth and mass-spring system $W_{\text{net}} = W_{ab,1} + W_{ba,2} = 0$.

We shall prove that if a force is conservative then the work done on a particle between two points a and b does **not** depend on the path.

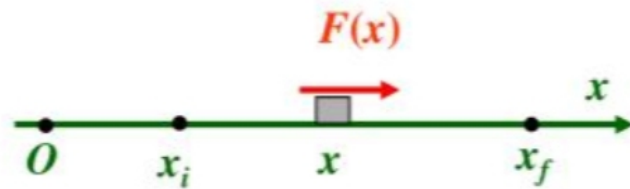
From fig. b we have: $W_{\text{net}} = W_{ab,1} + W_{ba,2} = 0 \rightarrow W_{ab,1} = -W_{ba,2}$ (eq. 1)

From fig.a we have $W_{ab,2} = -W_{ba,2}$ (eq. 2)

If we compare eq. 1 and eq. 2 we get:

$$W_{ab,1} = W_{ab,2}$$

(8-5)



Determining Potential Energy Values :

In this section we will discuss a method that can be used to determine the difference in potential energy ΔU of a conservative force F between points x_f and x_i on the x -axis if we know $F(x)$.

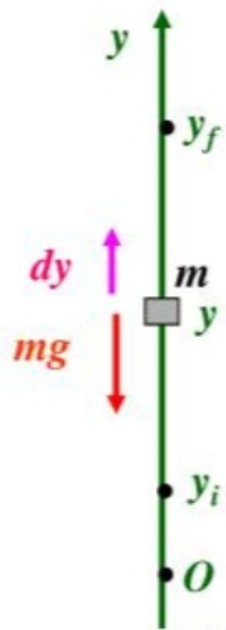
A conservative force F moves an object along the x -axis from an initial point x_i to a final point x_f . The work W that the force F does on the object is given by

$W = \int_{x_i}^{x_f} F(x)dx$. The corresponding change in potential energy ΔU was defined as

$\Delta U = -W$. Therefore the expression for ΔU becomes

$$\Delta U = - \int_{x_i}^{x_f} F(x)dx$$

(8-6)



Gravitational Potential Energy :

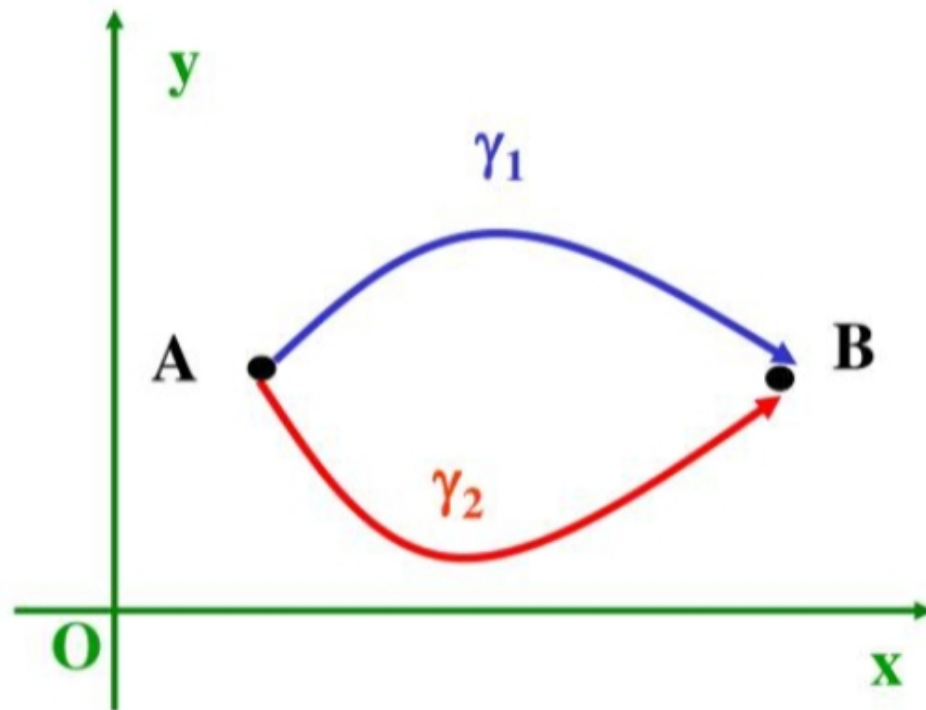
Consider a particle of mass m moving vertically along the y -axis from point y_i to point y_f . At the same time the gravitational force does work W on the particle, which changes the potential energy of the particle-Earth system. We use the result of the previous section to calculate ΔU .

$$\Delta U = - \int_{y_i}^{y_f} F(x) dy \quad F = -mg \quad \rightarrow \quad \Delta U = - \int_{y_i}^{y_f} (-mg) dy = mg \int_{y_i}^{y_f} dy = mg [y]_{y_i}^{y_f}$$

$\Delta U = mg(y_f - y_i) = mg\Delta y$. We assign the final point y_f to be the "generic" point y on the y -axis whose potential is $U(y)$ $\rightarrow U(y) - U_i = mg(y - y_i)$. Since only changes in the potential are physically meaningful, this allows us to define arbitrarily y_i and U_i . The most convenient choice is $y_i = 0$, $U_i = 0$. This particular choice gives

$$U(y) = mgy$$

(8-7)



Potential Energy U

U is defined for conservative forces **only**

Point A (x_A, y_A)

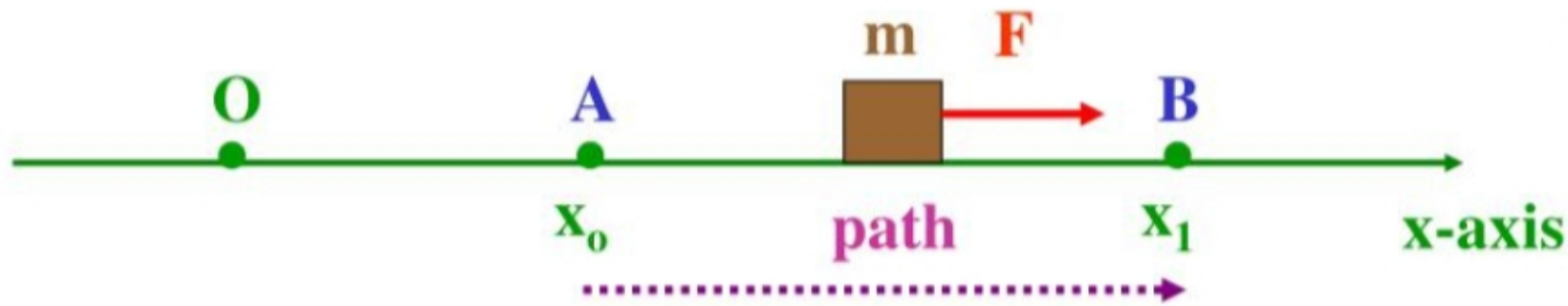
Point B (x_B, y_B)

W_{AB} = work done by a conservative force $\mathbf{F}$ from A to B

W_{AB} does **not** depend on the path taken. This can be expressed mathematically as follows: $W_{AB} = W(A,B)$

Also it can be written as: $W_{AB} = U(A) - U(B)$

U is the potential energy. **Units: J**

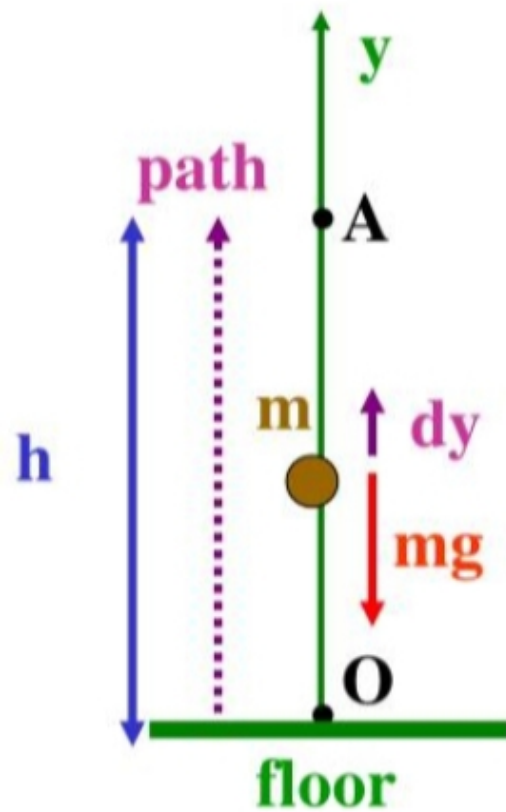


We assume that F depends on position i.e. $F = F(x)$

$$W_{AB} = U(x_0) - U(x_1) \quad \text{Also} \quad W_{AB} = \int_{x_0}^{x_1} F(x) dx \quad \rightarrow$$

$$U(x_1) - U(x_0) = - \int_{x_0}^{x_1} F(x) dx$$

Point A at x_0 is called the “**reference point**”. The choice of x_0 and $U(x_0)$ does not affect the result of our calculations. Thus we will choose x_0 and $U(x_0)$ for maximum convenience



Example: Potential energy of the gravitational force

Reference point: Point O at $y_0 = 0$

Point A at $y_1 = h$

$$U(y_1) - U(y_0) = -\int_0^h F_y dy = -\int_0^h (-mg) dy = mg \int_0^h dy$$

$$U(y_1) - U(y_0) = mg [y]_0^h = mgh$$

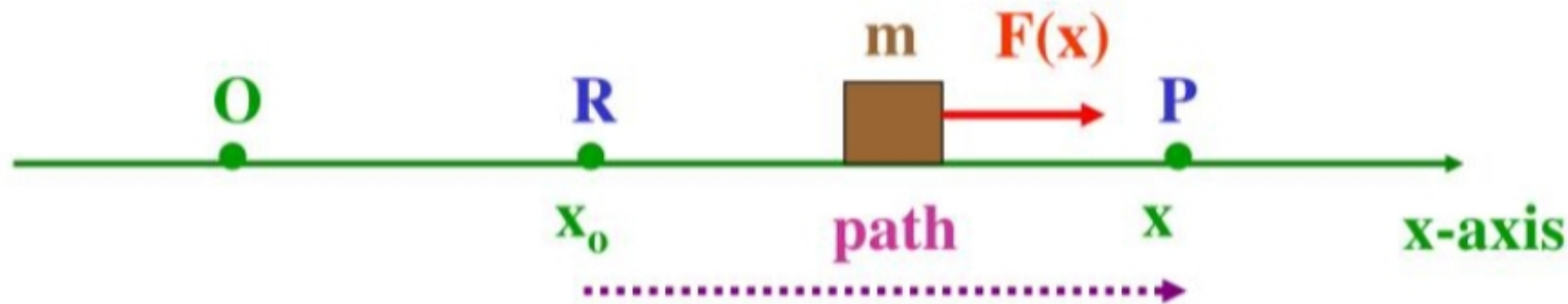
We choose: $U(y_0) = 0$

Thus we get :

$$U = mgh$$

Summary

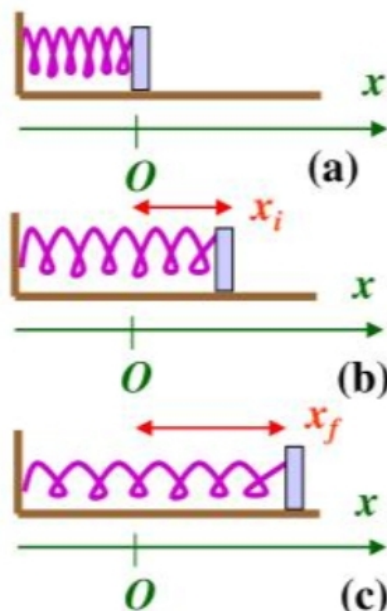
If we know $F(x)$ along the x -axis we can determine U at any point P using the following equation:



$$U(x) - U(x_0) = - \int_{x_0}^x F(x') dx'$$

The equation above gives $U(x)$ for every point P on the x -axis with coordinate x .

$$\mathbf{F(x) \rightarrow U(x)}$$



Potential Energy of a Spring :

Consider the block-mass system shown in the figure.

The block moves from point x_i to point x_f . At the same time the spring force does work W on the block, which changes the potential energy of the block-spring system by an amount

$$W = \int_{x_i}^{x_f} F(x)dx = \int_{x_i}^{x_f} -kx dx = -k \int_{x_i}^{x_f} x dx. \quad \Delta U = -W \rightarrow$$

$$\Delta U = k \left[\frac{x^2}{2} \right]_{x_i}^{x_f} = \frac{kx_f^2}{2} - \frac{kx_i^2}{2}. \quad \text{We assign the final point } x_f \text{ to be the "generic"}$$

point x on the x -axis whose potential is $U(x) \rightarrow U(x) - U_i = \frac{kx^2}{2} - \frac{kx_i^2}{2}.$

Since only changes in the potential are physically meaningful, this allows us to define arbitrarily x_i and U_i . The most convenient choice is

$x_i = 0$, $U_i = 0$. This particular choice gives

$$U = \frac{kx^2}{2} \quad (8-8)$$

Conservation of Mechanical Energy :

Mechanical energy of a system is defined as the sum of potential and kinetic energies $E_{\text{mech}} = K + U$. We assume that the system is isolated, i.e., no external forces change the energy of the system. We also assume that all the forces in the system are conservative. When an internal force does work W on an object of the system this changes the kinetic energy by $\Delta K = W$ (eq. 1). This amount of work also changes the potential energy of the system by an amount $\Delta U = -W$ (eq. 2).

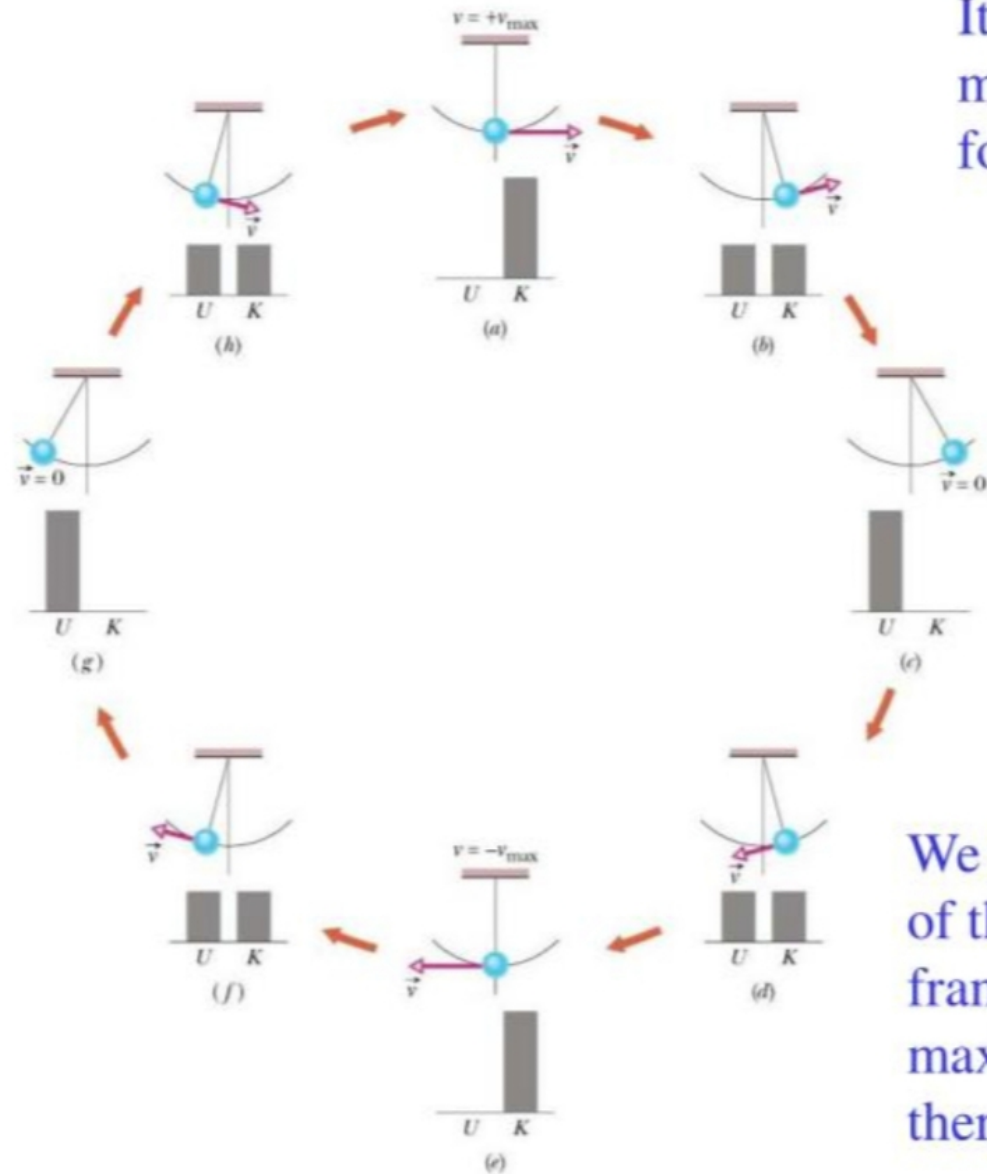
If we compare equations 1 and 2 we have: $\Delta K = -\Delta U \rightarrow K_2 - K_1 = -(U_2 - U_1) \rightarrow K_1 + U_1 = K_2 + U_2$. This equation is known as the **principle of conservation of mechanical energy**. It can be summarized as

$$\Delta E_{\text{mech}} = \Delta K + \Delta U = 0$$

For an isolated system in which the forces are a mixture of conservative and nonconservative forces the principle takes the following form: $\Delta E_{\text{mech}} = W_{nc}$

Here, W_{nc} is defined as the work of all the **nonconservative forces** of the system.

An example of the principle of conservation of mechanical energy is given in the figure. It consists of a pendulum bob of mass m moving under the action of the gravitational force.



The total mechanical energy of the bob-Earth system remains constant. As the pendulum swings, the total energy E is transferred back and forth between kinetic energy K of the bob and potential energy U of the bob-Earth system.

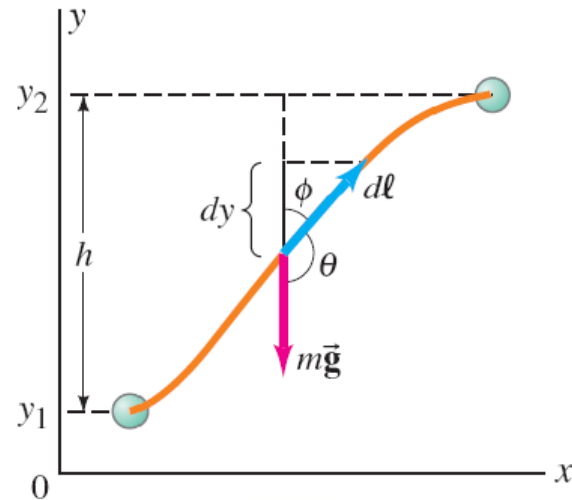
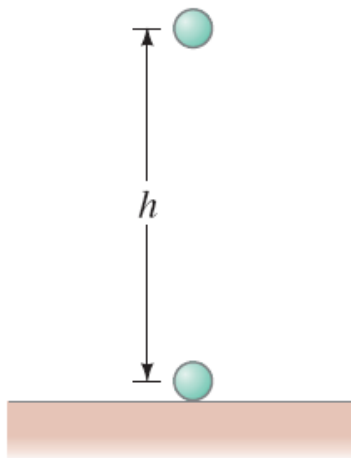
We assume that U is zero at the lowest point of the pendulum orbit. K is maximum in frames a and e (U is minimum there). U is maximum in frames c and g (K is minimum there).

Conservative and Nonconservative Forces

A force is conservative if:

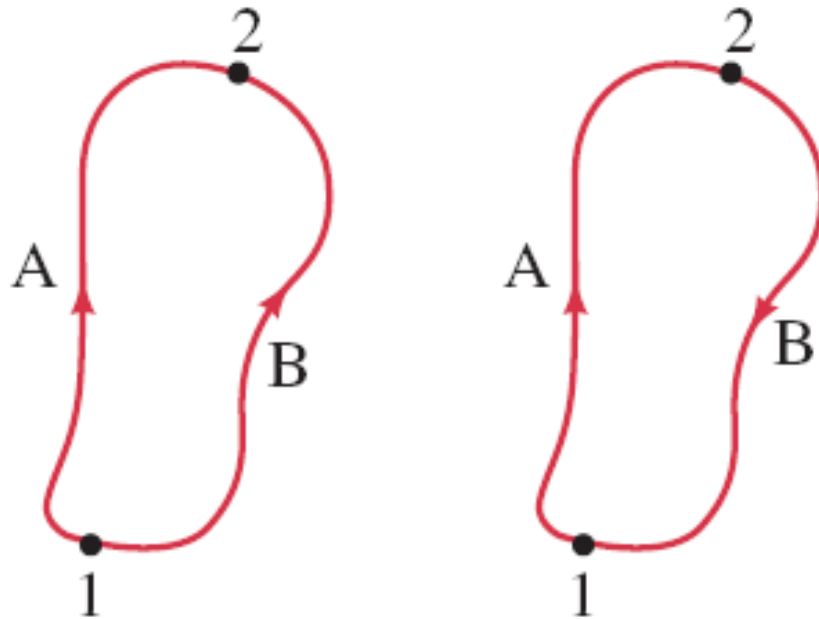
the work done by the force on an object moving from one point to another depends only on the initial and final positions of the object, and is independent of the particular path taken.

Example: gravitational force



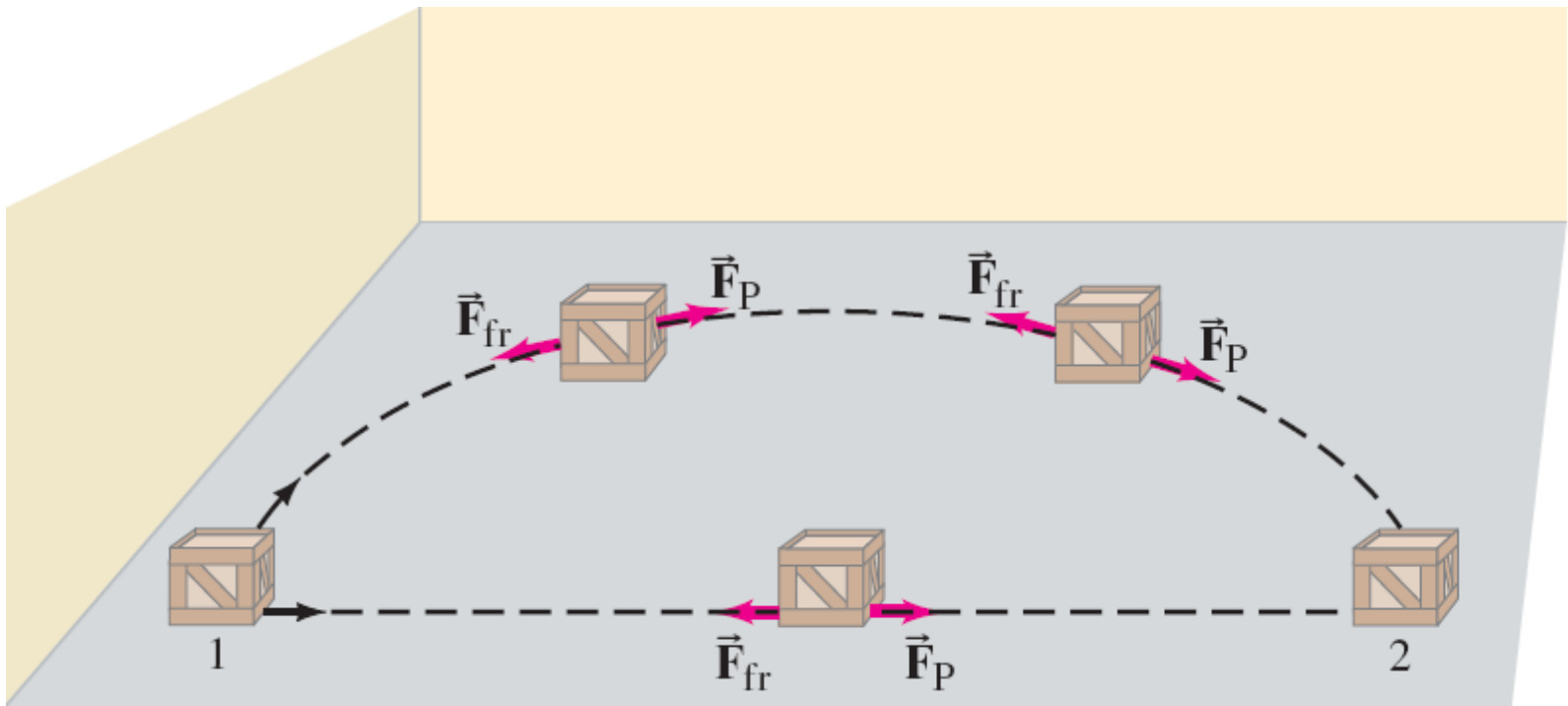
Another definition of a conservative force:

a force is conservative if the net work done by the force on an object moving around any closed path is zero.



Conservative and Nonconservative Forces

If **friction** is present, the work done depends not only on the starting and ending points, but also on the **path** taken. Frictional force is a **nonconservative force**.



Conservative and Nonconservative Forces

TABLE 8–1 Conservative and Nonconservative Forces

Conservative Forces	Nonconservative Forces
Gravitational	Friction
Elastic	Air resistance
Electric	Tension in cord
	Motor or rocket propulsion
	Push or pull by a person

Potential energy can be defined for conservative forces.

Gravitational Potential Energy

In raising a mass m from y_1 to y_2 by a height h , the work done by the external force is

$$\begin{aligned} W_{\text{ext}} &= \vec{\mathbf{F}}_{\text{ext}} \cdot \vec{\mathbf{d}} = mgh \cos 0^\circ \\ &= mgh = mg(y_2 - y_1) \end{aligned}$$

The gravitational potential energy is raised by

$$\Delta U_G = mg(y_2 - y_1) = mgh$$

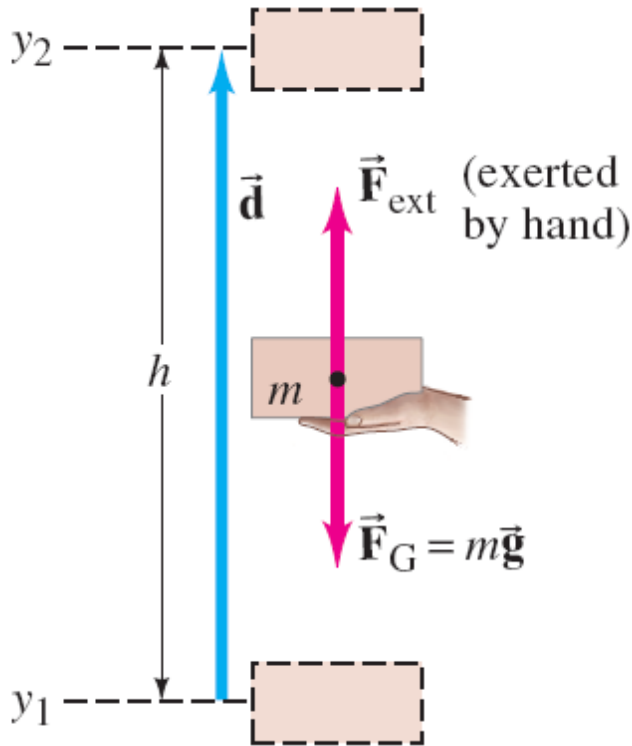
The work done by the gravitational force

$$W_G = m\vec{\mathbf{g}} \cdot \vec{\mathbf{d}} = -mg(y_2 - y_1) = -mgh$$

Therefore,

$$W_G = -\Delta U_G$$

The work done by the gravitational force is equal to the loss in potential energy.



General Definition of Potential Energy

For any conservative force $\vec{F}_C$

$$\Delta U = -W_C$$

i.e., the change in potential energy is equal to the negative work done by the conservative force.

Or, the work done by the conservative force is equal to the loss in potential energy. i.e.,

$$W_C = -\Delta U = -(U_{final} - U_{initial})$$

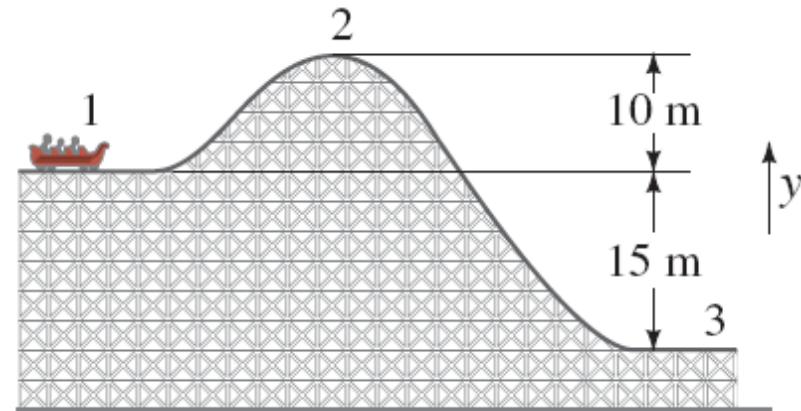
Note: we have defined the change in potential energy ΔU , not the potential energy itself. That is why we can arbitrary choose the zero point of potential energy U .

What is potential energy?

It's a position-determined capability of doing work.

Example: Potential energy changes for a roller coaster.

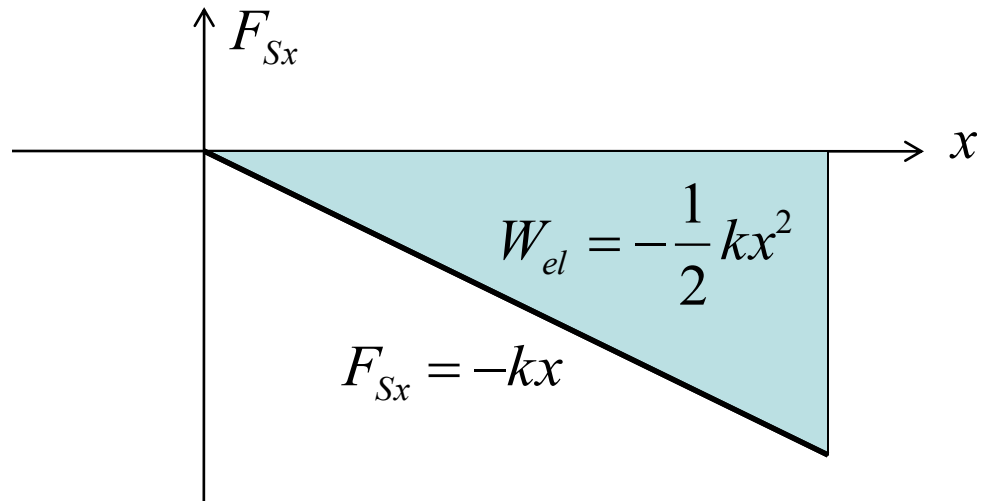
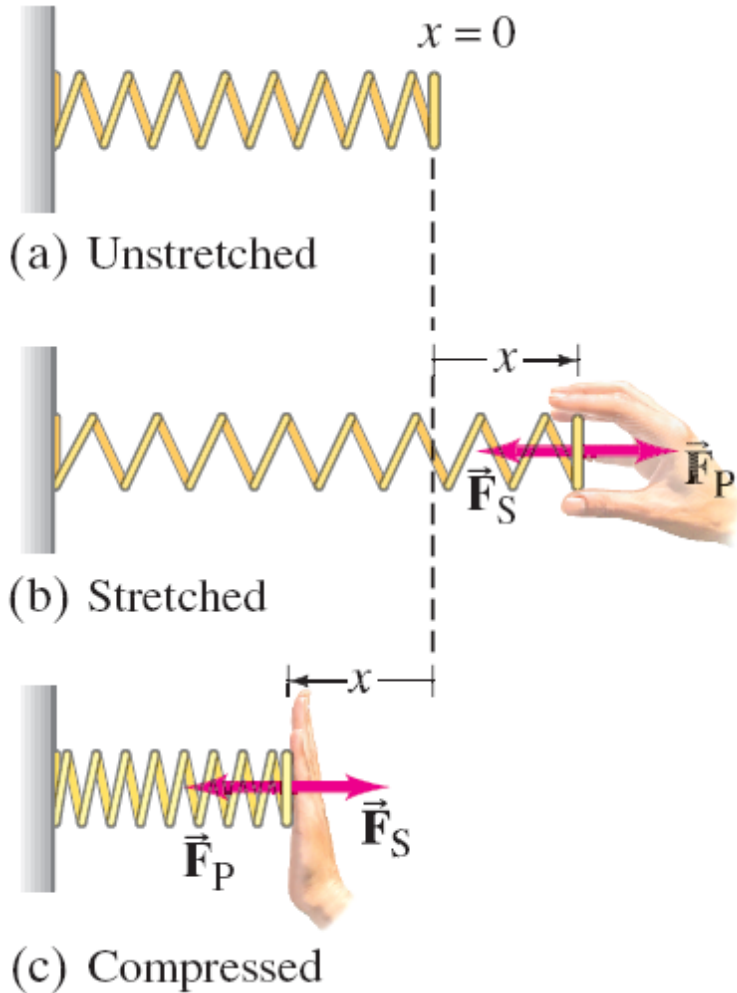
A 1000-kg roller-coaster car moves from point 1 to point 2 and then to point 3. (a) What is the gravitational potential energy at points 2 and 3 relative to point 1? That is, take $y = 0$ at point 1. (b) What is the change in potential energy when the car goes from point 2 to point 3? (c) Repeat parts (a) and (b), but take the reference point ($y = 0$) to be at point 3.



A spring has potential energy, called elastic potential energy.

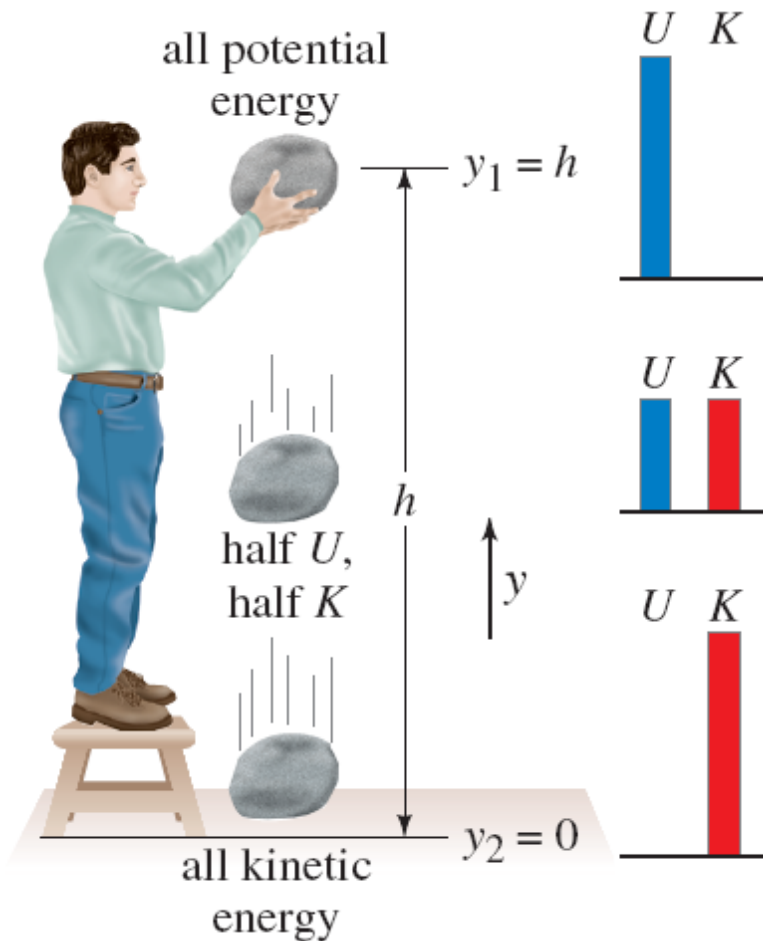
It is convenient to choose $x=0$ at the equilibrium position as well as the zero point of the elastic potential energy. Then if we stretch the spring by x , the elastic potential energy is

$$U_{el}(x) = U_{el}(x) - U_{el}(0) = -W_{el} = \frac{1}{2}kx^2$$



When we stretch (or compress) the spring by x , we raise the potential energy by $kx^2/2$.

Conservation of Mechanical Energy

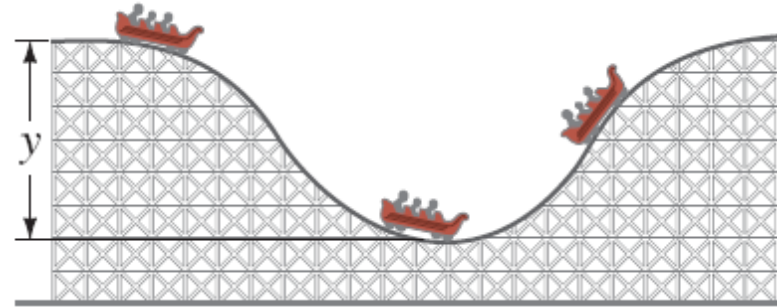


In the image on the left, the total mechanical energy at any point is:

$$\begin{aligned} E &= K + U \\ &= \frac{1}{2}mv^2 + mgy. \end{aligned}$$

Example: Roller-coaster car speed using energy conservation.

Assuming the height of the hill is 40 m, and the roller-coaster car starts from rest at the top, calculate (a) the speed of the roller-coaster car at the bottom of the hill, and (b) at what height it will have half this speed. Take $y = 0$ at the bottom of the hill.



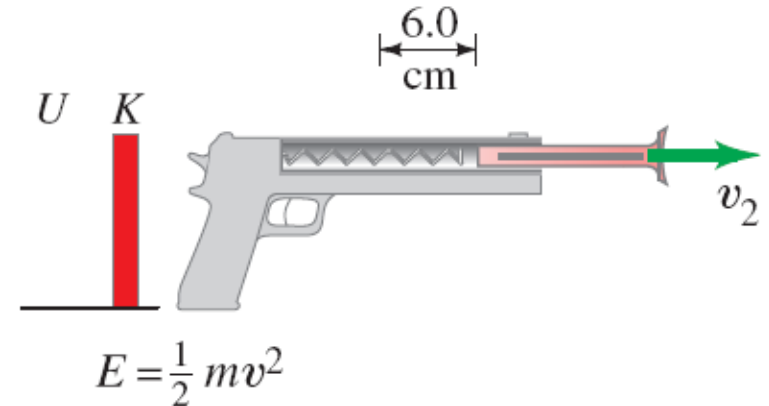
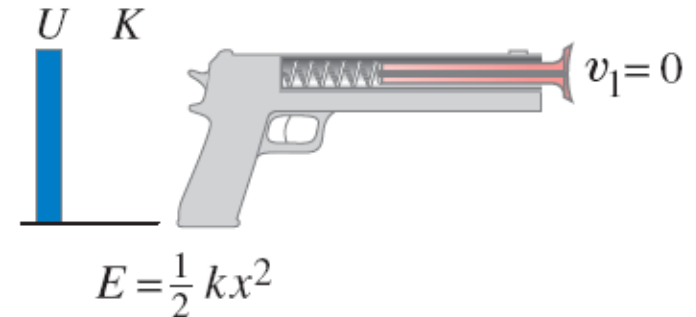
Conservation of Mechanical Energy

For an elastic force, conservation of energy tells us:

$$\frac{1}{2}mv_1^2 + \frac{1}{2}kx_1^2 = \frac{1}{2}mv_2^2 + \frac{1}{2}kx_2^2.$$

Example: Toy dart gun.

A dart of mass 0.100 kg is pressed against the spring of a toy dart gun. The spring (with spring stiffness constant $k = 250$ N/m and ignorable mass) is compressed 6.0 cm and released. If the dart detaches from the spring when the spring reaches its natural length ($x = 0$), what speed does the dart acquire?



The Law of Conservation of Energy

The law of conservation of energy is one of the most important principles in physics.

The total energy is neither increased nor decreased in any process. Energy can be transformed from one form to another, and transferred from one object to another, but the total amount remains constant.